# Gayathri Devi S

## 23MIA1019_BKanishk_23MIA1046_DeepakThirumal_v2

### Document Details

Submission ID
trn:oid:::3618:155980668

Submission Date
Oct 26, 2026, 6:27 PM GMT+0

Download Date
Oct 26, 2026, 6:51 PM GMT+0

File Name
23MIA1019_BKanishk_23MIA1046_DeepakThirumal_v2.docx

File Size
634.0 KB

8 Pages

3,667 Words

22,380 Characters

# 64% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely revised by AI.

**Caution: Review required.**

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

**36** AI-generated 64%
Likely generated or likely generated and revised by AI.

**Disclaimer**
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely human generated as AI generated and likely AI generated as human generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

# ParkSync: Duration-Aware and Vehicle-Aware Slot Allocation for Intelligent Multi-Level Parking with Digital Twin Integration

Deepak Thirumal
VIT Chennai
Chennai, India

B. Kanishk
VIT Chennai
Chennai, India

**Abstract:**

**Parking Management systems traditionally focus on detecting vacant spaces while giving limited consideration to the expected duration of occupancy and the other requirements of different vehicle types. ParkSync addresses this problem through a duration aware and vehicle aware parking slot allocation strategy for multi-level parking facilities. The proposed approach predicts the expected parking duration of an arriving vehicle using a Random Forest regression model and converts the predicted dwell time into a normalized parking depth requirement. This requirement is matched against a facility representation incorporating accessibility rank, zone depth, and floor depth. Vehicle specific constraints are incorporated through EV compatibility handling, allowing electric vehicles to be preferentially assigned to compatible EV slots when such slots are available. Experiments were conducted using 6,000 synthetically generated parking records and a deterministic 24 slot, two floor parking facility. ParkSync is compared with nearest slot and duration only allocation strategies**

***Keywords:*** Smart parking, parking duration prediction, slot allocation, multi-level parking, electric vehicles, Digital Twin, machine learning, intelligent transportation systems.

## I. INTRODUCTION

Urban parking facilities increasingly face challenges caused by limited capacity, unpredictable parking durations. Traditional parking systems commonly prioritize finding an available space or assigning the nearest available space. Although proximity based allocation can reduce the immediate travel distance of a vehicle, it does not take into account the duration for which a vehicle is expected to occupy a slot. Consequently, short stay and long stay vehicles may compete for the same highly accessible spaces even though their operational requirements differ.

The expected parking duration provides useful information for improving spatial utilization. A vehicle expected to remain for only a short period can benefit from a highly accessible parking position because the slot is likely to become available relatively soon. While, a vehicle expected to remain for several hours can be placed farther from the primary access routes without imposing the same repeated accessibility cost.

Existing research has investigated real time parking reservation, dynamic parking allocation, parking duration prediction, EV parking allocation, and Digital Twin based parking systems. However, these research directions are generally addressed through separate formulations. The specific problem considered in this work is how a predicted parking dwell duration can be explicitly mapped to the accessibility structure of a multi-level parking facility while simultaneously accounting for vehicle specific slot compatibility.

ParkSync addresses this problem through a duration to depth slot allocation formulation. A machine learning model predicts parking duration. The predicted duration is normalized into a dwell factor. Each parking slot is represented using a depth score derived from its accessibility rank, zone accessibility, and floor position. The allocation engine then maximizes the alignment between the predicted dwell requirement and the slot depth while incorporating EV compatibility.

The principal contribution of ParkSync is therefore the allocation mechanism rather than the individual technologies used to implement it. Specifically, the work contributes:

1) A duration to depth formulation that converts predicted parking dwell time into a normalized spatial placement requirement.

2) A rule based multi-level facility representation that combines accessibility rank, zone depth, and floor depth.

3) An integrated vehicle aware allocation mechanism that incorporates EV compatibility into duration aware slot assignment.

4) A controlled experimental comparison against representative nearest slot and duration only allocation strategies using an identical facility and arrival stream.

## II. RELATED WORK

### *A. Intelligent Parking Allocation*

Research on intelligent parking has increasingly moved beyond simple occupancy detection toward dynamic reservation and allocation. Bai et al. proposed a two-stage approximation approach for real time parking reservations under stochastic requests and reusable parking resources, explicitly considering varying parking durations and uncertain requests. Wang et al. developed a two-stage reservation and allocation approach considering parking unpunctuality, using probabilistic models and rolling horizon allocation. These studies demonstrate the importance of dynamic allocation when arrival and parking duration uncertainty affect resource availability.

Optimization based approaches have also been investigated for dynamic parking space allocation. Such approaches formulate parking assignment as an optimization problem subject to temporal, spatial, or resource constraints. These methods provide evidence that parking allocation can be treated as a resource assignment problem rather than merely a vacancy detection problem.

ParkSync converts predicted dwell duration into a spatial depth requirement for a multi-level facility and incorporating vehicle specific EV compatibility within the same allocation decision.

### *B. Parking Duration Prediction*

Parking duration prediction has been studied as a mechanism for improving parking management. Recent work on parking duration prediction for shopping mall parking facilities has evaluated multiple regression and classification approaches and examined how duration predictions can support parking space assignment.

Phipps et al. investigated uncertainty quantification for parking duration prediction in EV smart charging applications, demonstrating the relevance of expected parking duration to subsequent resource management decisions.

These studies establish that parking duration can be predicted from contextual information and can provide useful information for downstream parking or energy decisions. In ParkSync, the predicted duration is used as the input to a specific duration to depth parking allocation mechanism.

### *C. EV Aware Parking Allocation*

EV specific parking management introduces additional constraints because EVs may require charging compatible spaces. Sadreddini et al. proposed a decision based multicriteria reservation system for EV parking lots considering factors such as user behavior, parking availability, state of charge, and usage history.

These studies demonstrate the importance of incorporating EV requirements into parking or charging decisions. In ParkSync, EV compatibility is integrated into the same slot selection mechanism that matches predicted parking duration with the spatial depth of a multi-level facility.

### *D. Digital Twin Based Parking Systems*

Digital Twin technologies have increasingly been applied to intelligent transportation and parking environments. Research has investigated Digital Twin architectures combined with optimization, multi objective decision making, vehicular sensing. Such systems demonstrate the value of maintaining a digital representation of parking infrastructure and its operational state.

In ParkSync, the Digital Twin represents the operational parking state, including floors, zones, slots, occupancy, assigned vehicle information, EV capability, and spatial organization.

### *E. Research Gap*

The existing literature demonstrates that parking duration prediction, dynamic parking allocation, EV aware parking management, and Digital Twin based parking systems are each established research directions. The remaining problem addressed by ParkSync is the specific allocation formulation that combines these decision requirements at the slot selection level.

While existing studies have explored duration aware parking allocation and EV specific parking

management independently, there remains a need for an allocation mechanism that explicitly maps predicted dwell duration to the accessibility structure of a multi-level parking facility while simultaneously enforcing vehicle specific slot compatibility.

## III. PROPOSED PARKSYNC ARCHITECTURE

### *A. System Overview*

ParkSync is implemented as a layered intelligent parking system consisting of a client layer, service layer, machine learning and computer vision components, state management infrastructure, and a Digital Twin visualization layer.

The user interacts with the parking system through a React based client. The parking request contains vehicle related and session related information.

The FastAPI backend provides the main service layer. Parking session management, duration prediction, slot allocation, computer vision services, and event handling are integrated through this layer.

The architecture separates persistent data management from low latency operational state. MongoDB is used for persistent application and parking session data, while Redis maintains low latency operational state. WebSocket communication is used to broadcast parking state changes to connected clients.

The Three.js Digital Twin receives these state updates and renders the corresponding parking facility state.

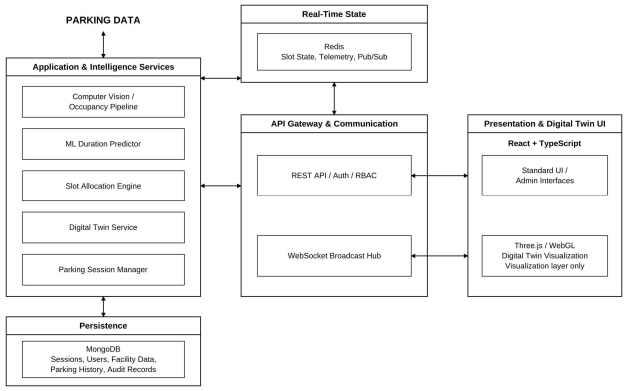



Fig. 1. ParkSync system architecture and operational data flow.

### *B. Digital Twin*

The ParkSync Digital Twin represents the modeled operational state of the parking facility. It includes two floors, four zones, 24 parking slots, occupancy state, assigned vehicles, EV capability, and spatial coordinates.

When a parking state updation occurs, the operational state is updated and propagated through the state management layer. Redis provides low latency state access, while persistent information can be stored in MongoDB. WebSocket events propagate state changes to the client, where the Three.js representation is updated.

The Digital Twin is therefore treated as an operational software representation of the parking environment rather than merely a static 3 dimensional visualization.

### *C. Backend and State Management*

FastAPI provides the service layer connecting client requests to parking session management, machine learning inference, allocation logic, and state management services.

The system is implemented primarily using open-source software components, including Python, FastAPI, React, Three.js, OpenCV, scikit-learn, Redis, MongoDB, and WebSocket based communication.

### *D. Parking Session Management*

Parking sessions contain vehicle and duration related information used by the allocation system. A parking request is processed through the duration prediction component before being passed to the allocation engine.

The predicted duration is used when selecting a slot, while the actual duration is used during simulation to determine when the occupied slot becomes available again.

## IV. METHODOLOGY

### *A. Synthetic Dataset*

The duration prediction experiment uses exactly 6,000 synthetically generated parking records.

Each record contains: expected_duration_minutes, purpose, arrival_hour, day_of_week, is_weekend, vehicle_type, and actual_duration_minutes.

The purpose categories are Shopping, Office, College, Restaurant, Appointment, Personal, and Other. Vehicle categories are Sedan, SUV, EV, Compact, and Motorcycle.

The dataset generator uses fixed random seeds of 42 for NumPy and Python's random module to improve reproducibility.

Actual parking duration is generated using purpose specific normal distributions and bounded to the interval of 10 to 600 minutes. Expected duration is generated from actual duration using an estimation ratio with mean approximately 0.92 and standard deviation approximately 0.12, together with rounding noise.

TABLE I
DATASET FEATURES

| Feature | Description |
|---|---|
| expected_duration_minutes | Initial expected parking duration |
| purpose | Parking purpose category |
| arrival_hour | Hour of vehicle arrival |
| day_of_week | Day index |
| is_weekend | Weekend indicator |
| vehicle_type | Vehicle category |
| actual_duration_minutes | Generated actual parking duration used as target |

*B. Duration Prediction Model*

The duration prediction component uses a scikit-learn Pipeline containing preprocessing and a RandomForestRegressor.

Categorical input features are purpose and vehicle_type. Numerical input features are expected_duration_minutes, arrival_hour, day_of_week, and is_weekend. The prediction target is actual_duration_minutes.

The dataset is divided into training and testing subsets using an 80/20 split with random_state = 42.

TABLE II
DURATION PREDICTION MODEL CONFIGURATION

| Parameter | Value |
|---|---|
| Model | RandomForestRegressor |
| Number of trees | 120 |
| Maximum depth | 14 |
| Minimum samples split | 4 |
| Minimum samples leaf | 2 |
| Train/test split | 80/20 |
| Random state | 42 |
| Target | actual_duration_minutes |

The resulting model achieves: MAE = 16.96 minutes, RMSE = 26.33 minutes, $R^2 = 0.9763$.

*C. Multi-Level Facility Representation*

The experimental facility contains two floors, four zones, and 24 parking slots.

TABLE III
PARKING FACILITY CONFIGURATION

| Property | Configuration |
|---|---|
| Floors | 2 |
| Zones | 4 |
| Slots | 24 |
| Accessibility ranks | 1–24 |
| Prime half | Ranks 1–12 |
| Floor 1 | Slots 1–12 |
| Floor 2 | Slots 13–24 |
| EV slots | Slots 7–12 |

Accessibility rank equals slot ID, with rank 1 representing the most accessible slot and rank 24 representing the deepest slot.

*D. Parking Depth Model*

ParkSync represents the spatial accessibility of each slot through three components: accessibility rank, zone depth, and floor depth.

The rank factor is:

$$rank_factor = clip((rank - 1)/(R_{max} - 1), 0, 1) \quad (1)$$

For the 24-slot facility, rank 1 maps to 0 and rank 24 maps to 1.

The zone depth is defined as:

$$zone_depth = 1 - accessibility$$

The predefined zone accessibility values are: Zone 101 = 0.95, Zone 102 = 0.85, Zone 201 = 0.45, Zone 202 = 0.35.

The final slot depth is:

$$S_depth = 0.45\ rank_factor + 0.35\ zone_depth + 0.20\ floor_depth$$

TABLE IV

PARKING DEPTH AND ALLOCATION PARAMETERS

| Parameter | Value / Definition |
|---|---|
| Minimum duration reference | 15 min |
| Maximum duration reference | 480 min |
| Rank factor weight | 0.45 |
| Zone depth weight | 0.35 |
| Floor depth weight | 0.20 |
| Floor 1 depth | 0.00 |
| Floor 2 depth | 0.40 |
| Zone 101 depth | 0.05 |
| Zone 102 depth | 0.15 |
| Zone 201 depth | 0.55 |
| Zone 202 depth | 0.65 |
| Duration alignment weight | 0.75 |
| EV compatibility weight | 0.25 |

*E. Duration to Depth Matching*

Let d denote the predicted parking duration in minutes. The dwell factor is calculated as:

$$dwell_factor = clip((d - 15)/(480 - 15), 0, 1)$$

Durations below 15 minutes are mapped to zero and durations above 480 minutes are mapped to one.

The alignment score between the predicted dwell requirement and a slot's depth is:

$$A = max(0, 1 - |dwell_factor - S_depth|)$$

A larger alignment score indicates that the slot depth is better matched to the expected duration of the vehicle.

*F. EV Compatibility*

ParkSync incorporates EV compatibility into the slot score. For an EV vehicle, E = 1.00 for an EV compatible slot, E = 0.05 for a non-EV slot. For a non-EV vehicle, E = 0.85 for an EV slot, E = 1.00 for a non-EV slot. The final composite score is:

$$S = 0.75A + 0.25E$$

A critical implementation detail is that EV compatibility is not purely represented by the soft EV factor. If an EV compatible slot exists, the candidate set for an EV vehicle is first restricted to EV compatible slots. The composite score is then calculated among the eligible candidates.

If no EV compatible slot is available, the allocation engine falls back to the complete set of available slots.

*G. Final Slot Allocation Algorithm*

ALGORITHM 1
PARKSYNC DURATION AND VEHICLE AWARE SLOT ALLOCATION

```
Input: Available slots, predicted
duration, vehicle type, EV
requirement.
1. Convert predicted duration to
minutes.
2. Clamp predicted duration to a
minimum of 5 minutes.
3. Calculate dwell_factor.
4. Determine the available slot
ranks.
5. Calculate rank_factor for each
candidate slot.
6. Calculate zone_depth and
floor_depth.
7. Calculate slot depth.
8. Calculate duration/depth
alignment score.
9. If EV compatibility is required
and EV slots are available, restrict
candidates to EV slots.
10. Calculate the EV compatibility
factor.
11. Calculate the composite score
using 0.75 alignment weight and 0.25
EV weight.
12. Apply the duration dependent
rank tie breaker.
13. Select the highest scoring slot.
14. Return the selected slot.
```

*H. Baseline Strategies*

Three allocation policies are evaluated. Nearest slot allocation selects the available slot with the lowest accessibility rank.

Duration only allocation uses the duration to depth alignment mechanism but does not incorporate EV aware scoring.

ParkSync uses duration prediction, facility depth representation, EV compatibility filtering, and the 75/25 duration alignment and EV compatibility scoring mechanism.

## V. EXPERIMENTAL SETUP

### *A. Simulation Environment*

The same 6,000 records are evaluated under all three allocation policies. Each allocation policy operates on the same 24 slot facility. Predicted parking duration is used for slot assignment. Actual parking duration is used to determine slot release.

This design separates prediction based allocation from occupancy release behavior while ensuring that all policies experience identical arrival and capacity conditions.

### *B. Stay Categories*

Parking sessions are divided into three categories according to predicted duration. Short stay: d <= 120 minutes. Medium stay: 120 < d <= 180 minutes. Long stay: d > 180 minutes. The prime half of the facility consists of ranks 1–12.

### *C. Evaluation Metrics*

The following metrics are used:

Mean short stay rank: The average accessibility rank assigned to short stay vehicles. Lower values indicate more accessible placement.

Short stay prime half allocation: The percentage of short stay sessions assigned to ranks 1-12.

Mean long stay rank: The average accessibility rank assigned to long stay vehicles.

EV sessions assigned to EV slots: The percentage of accepted EV sessions assigned to EV compatible slots.

Acceptance rate: The percentage of the 6,000 sessions that obtain a slot.

## VI. RESULTS AND DISCUSSION

### *A. Overall Comparison*

The complete comparison is presented in Table V.

TABLE V
COMPARISON OF PARKING ALLOCATION POLICIES

| **Policy** | **Mean Short Stay Rank** | **Short Stay Rank Reduction vs Nearest** | **Short Stay Prime Half Allocation** | **Mean Long Stay Rank** | **EV Sessions Assigned to EV Slots** | **Acceptance Rate** |
|---|---|---|---|---|---|---|
| Nearest slot | 12.2746 | 0.00% | 50.16% | 12.4501 | 23.88% | 63.52% |
| Duration only | 9.6935 | 21.03% | 68.59% | 13.3540 | 28.87% | 63.52% |
| Park Sync | 9.9760 | 18.73% | 66.67% | 13.2984 | 35.96% | 63.52% |

The results show that both duration aware strategies substantially change the spatial distribution of short stay vehicles compared with nearest slot allocation.

ParkSync reduces the mean short stay rank from 12.2746 under nearest slot allocation to 9.9760. This corresponds to an 18.73% reduction. The proportion of short stay sessions assigned to the prime half increases from 50.16% to 66.67%, representing an improvement of approximately 16.50 percentage points.

Duration only allocation produces a slightly lower mean short stay rank of 9.6935 and a higher prime half allocation of 68.59%. This indicates that duration to depth matching alone can produce slightly stronger pure spatial placement for short stay vehicles.

However, ParkSync provides a substantially stronger EV compatibility result. The proportion of accepted EV sessions assigned to EV compatible slots reaches 35.96%, compared with 28.87% for duration only allocation and 23.88% for nearest slot allocation.

Relative to nearest slot allocation, ParkSync therefore increases EV to EV slot assignment by approximately 12.08 percentage points. Relative to duration only allocation, the improvement is approximately 7.09 percentage points.

### *B. Short Stay Placement*

For short stay vehicles, nearest slot allocation produces a mean rank of 12.2746. Duration only

allocation reduces this to 9.6935, while ParkSync produces a mean rank of 9.9760.

The corresponding prime half allocation rates are 50.16%, 68.59%, and 66.67% for nearest slot, duration only, and ParkSync, respectively.

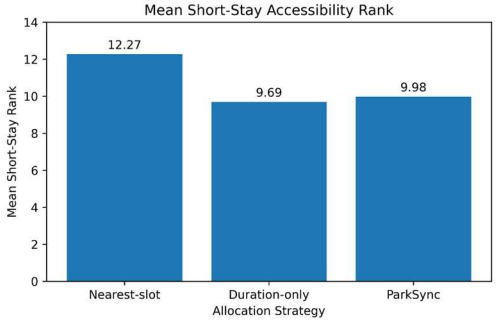



Fig. 2. Mean accessibility rank for short stay vehicles.

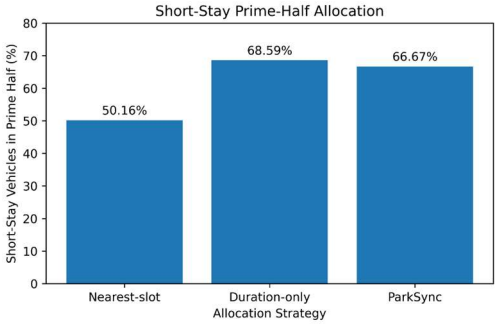



Fig. 3. Short stay sessions assigned to the prime half of the facility.

### *C. Long Stay Placement*

For long stay vehicles, the mean rank is 12.4501 for nearest slot allocation, 13.3540 for duration only allocation, and 13.2984 for ParkSync. The higher rank under duration aware approaches indicates that longer stay vehicles are more frequently assigned toward deeper regions of the facility. ParkSync produces a long stay mean rank of 13.2984, closely following duration only allocation at 13.3540.

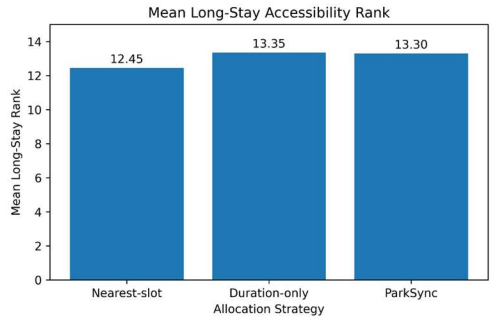



Fig. 4. Mean accessibility rank for long stay vehicles.

### *D. EV Compatibility*

The EV allocation results demonstrate the additional effect of vehicle aware slot handling. Nearest slot allocation assigns 23.88% of accepted EV sessions to EV compatible slots. Duration only allocation increases this value to 28.87%. ParkSync reaches 35.96%.

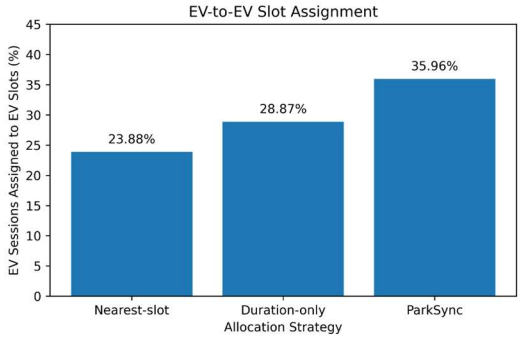



Fig. 5. EV sessions assigned to EV compatible parking slots.

This result reflects the explicit EV compatibility handling in ParkSync. When an EV compatible slot is available, the candidate set is restricted to compatible slots before the composite duration/EV score is applied.

### *E. Acceptance Rate and Capacity*

All three policies accept exactly 3,811 sessions and reject 2,189 sessions. The resulting acceptance rate is 63.52% for every policy.

This equality is expected because the simulation uses the same 24 slot capacity, identical arrival sequence, and actual duration based slot release for all allocation strategies. The allocation scoring policy changes which slot is selected but does not alter the underlying occupancy duration or facility capacity.

## VII. LIMITATIONS

Several limitations should be considered when interpreting the results.

First, the duration dataset contains 6,000 synthetically generated records rather than observations from a real parking facility. The strong duration prediction $R^2$ therefore reflects the structured synthetic relationships used to generate the data and should not be interpreted as evidence of equivalent real world predictive accuracy.

Second, the facility depth representation is rule based. Zone accessibility values, floor depth penalties, and weighting coefficients are predefined design

parameters rather than a physically measured distance or accessibility reduction.

Third, the Digital Twin is evaluated primarily as an operational software representation of the modeled parking facility. Full synchronization with a physical parking facility has not been experimentally validated.

Future experiments should use real parking duration observations, real facility geometry, measured accessibility information and uncertainty aware duration prediction.

## VIII. CONCLUSION AND FUTURE WORK

ParkSync, a duration aware and vehicle aware parking slot allocation strategy for multi-level parking facilities. The central mechanism transforms predicted parking dwell time into a normalized spatial requirement and matches that requirement against a facility depth representation based on accessibility rank, zone depth, and floor depth. EV compatibility is incorporated into the same allocation decision through compatibility aware candidate selection and scoring.

Future work will focus on replacing the synthetic duration data with real parking observations, calibrating the facility depth model using measured geometry and accessibility, incorporating uncertainty into duration prediction, and extending the system toward coordinated charging, dynamic pricing, larger parking facilities, and real world Digital Twin synchronization.

## REFERENCES

[1] M. Bai, P. Yan, Z. Chen, X. Cai, and X. T. R. Kong, “Two-stage approximation allocation approach for real-time parking reservations considering stochastic requests and reusable resources,” Advanced Engineering Informatics, vol. 59, Art. no. 102251, 2024, doi: 10.1016/j.aei.2023.102251.

[2] X. Wang, M. Xu, H. Xiao, and Q. Yang, “A two-stage reservation and allocation approach for smart parking system considering parking unpunctuality,” Transportation Research Part C, Art. no. 105304, 2025, doi: 10.1016/j.trc.2025.105304.

[3] “Optimizing Parking Efficiency Using Parking Duration Prediction: A Case Study of a Shopping Mall Parking Facility,” Applied Sciences, vol. 16, no. 13, Art. no. 6704, 2026.

[4] V. M. de Lira, F. Pallonetto, L. Gabrielli, and C. Renso, “Predicting vehicles parking behaviour in shared premises for aggregated EV electricity demand response programs,” arXiv preprint arXiv:2109.09666, 2021.

[5] K. Phipps, K. Schwenk, B. Briegel, R. Mikut, and V. Hagenmeyer, “Customized uncertainty quantification of parking duration predictions for EV smart charging,” IEEE Internet of Things Journal, vol. 10, no. 23, pp. 20649–20661, 2023, doi: 10.1109/JIOT.2023.3299201.

[6] Z. Sadreddini, S. Guner, O. Erdinc, et al., “Design of a decision-based multicriteria reservation system for the EV parking lot,” IEEE Transactions on Transportation Electrification, vol. 7, no. 4, pp. 2429–2438, 2021, doi: 10.1109/TTE.2021.3067953.

[7] S. Bilotta, L. A. Ipsaro Palesi, and P. Nesi, “Predicting free parking slots via deep learning in short-mid terms explaining temporal impact of features,” IEEE Access, vol. 11, pp. 101678–101693, 2023, doi: 10.1109/ACCESS.2023.3314660.

[8] “Constraint Optimization Model for Dynamic Parking Space Allocation,” Sensors, vol. 24, no. 12, Art. no. 3988, 2024.

[9] O. M. Abdelwahab, A. A. Shalaby, and M. F. Shaaban, “An optimal resource allocation for future parking lots with charger assignment considering uncertainties,” Electric Power Systems Research, vol. 200, Art. no. 107455, 2021, doi: 10.1016/j.epsr.2021.107455.

[10] “A Rolling-Horizon Framework for Managing Shared Parking and Electric Vehicle Charging,” Sustainable Cities and Society, vol. 98, Art. no. 104810, 2023.

[11] “A multi objective optimization framework for smart parking using digital twin pareto front MDP and PSO for smart cities,” Scientific Reports, vol. 15, Art. no. 7783, 2025.